$$\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \qquad (3.17)$$

**Exercise 3.2:** Prove that Eq. 3.16 is the unique solution to Eqs. 3.14 and 3.15.

This is our very first example of a quantum mechanical operator. Let's summarize what went into it. First, some experimental data: there are certain states that we called $|u\rangle$ and $|d\rangle$, in which the measurement of $\sigma_z$ gives unambiguous results ±1. Next, the principles told us that $|u\rangle$ and $|d\rangle$ are orthogonal and are eigenvectors of a linear operator $\sigma_z$. Finally, we learned from the principles that the corresponding eigenvalues are the observed (or measured) values, again ±1. That's all it takes to derive Eq. 3.17.

Can we do the same for the other two components of spin, $\sigma_x$ and $\sigma_y$? Yes, we can.[3] The eigenvectors of $\sigma_x$ are $|r\rangle$ and $|l\rangle$, with eigenvalues +1 and −1 respectively. In equation form,

$$\begin{aligned} \sigma_x|r\rangle &= |r\rangle \\ \sigma_x|l\rangle &= -|l\rangle. \end{aligned} \qquad (3.18)$$

Recall that $|r\rangle$ and $|l\rangle$ are linear superpositions of $|u\rangle$ and $|d\rangle$:

$$\begin{aligned} |r\rangle &= \frac{1}{\sqrt{2}}|u\rangle + \frac{1}{\sqrt{2}}|d\rangle \\ |l\rangle &= \frac{1}{\sqrt{2}}|u\rangle - \frac{1}{\sqrt{2}}|d\rangle. \end{aligned} \qquad (3.19)$$